Glints of Gold or Troubling Waters? Can a School of Merged Monolingual Goldfish Models Swim in Bilingual Seas?

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Abstract

Multilingual capabilities in language models can arise through joint pretraining, but recent work has identified a potential "curse of bilinguality" – adding a second language can hurt performance. Model merging, which combines independently trained monolingual models without further joint training, offers an alternative, though its effectiveness in small, data-constrained models is unclear. Using the Goldfish family of small monolingual models (Chang et al., 2024), we uniformly compare bilingual pretraining and merging to monolingual pretraining under identically controlled conditions in data-constrained evaluation settings. Simultaneous bilingual pretraining preserves monolingual capabilities and enables cross-lingual transfer, while sequential pretraining and naive model merging can impair monolingual capabilities in data-constrained, small model regimes.

`https://github.com/suchirsalhan/school-of-goldfish` (CC BY 4.0)

`https://huggingface.co/collections/suchirsalhan/school-of-goldfish` (CC BY 4.0)

1 Introduction

Jointly pretraining on multilingual data allows language models to learn many languages at the same time (Pires et al., 2019; Conneau et al., 2020b,a; Xue et al., 2021). This joint training has been argued to induce a shared representational space through joint exposure to multiple languages (Chang et al., 2022), enabling cross-lingual transfer and generalisation. More recently, however, multilingual capabilities have also been created through **model merging** (Wang et al., 2025; Ahmadian et al., 2024; Khairi et al., 2025). Model merging refers to a family of techniques that combine multiple independently trained models directly in parameter space, without additional joint training on the original data (Polyak and Juditsky (1992); Yadav et al. (2023, 2024) i.a.). In multilingual language modelling, this typically involves training separate monolingual models and integrating them via operations such as parameter averaging, linear interpolation, or task arithmetic, yielding a single model intended to support multiple languages while preserving competencies learned in isolation.

Despite promising results, it remains unclear under what conditions model merging is a viable alternative to multilingual or bilingual pretraining. Prior work has largely focused on large-scale models or high-resource settings (Ahmadian et al., 2024), or on improving performance for specific multilingual or code-mixed tasks (Bandarkar and Peng, 2025; Kodali et al., 2025), leaving *data-constrained, small-capacity regimes* largely unexplored. Furthermore, Zhou and Matusevych (2025) identify a **"curse of bilinguality"**, a form of negative cross-lingual transfer where adding a second language can degrade performance in one language compared to a monolingual model. To date, these effects have not been systematically investigated at the same scale across monolingual, bilingual pretraining, and model merging settings, nor has research examined how multilinguality ac-

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quired through merging differs from that acquired through joint pretraining, particularly in terms of linguistic knowledge encoding, integration, and transfer. These gaps hinder theory-driven multilingual model development (Diehl Martinez et al., 2025; Africa et al., 2025) and highlight the need for evaluation practices appropriate for low-data regimes, which are critical for long-tail languages.

In this paper, we explore on data-constrained **bilingual language models** to address two central research questions: (1) How do bilingual pretraining and bilingual model merging compare to monolingual pretraining in low-data, small-capacity settings? (2) How does linguistic knowledge differ when acquired via joint pretraining versus post-hoc merging? We systematically evaluate monolingual, bilingual, and merged models while controlling for model architecture, token budget, and language exposure, isolating the effects of training regime and integration strategy. Using inter-model comparisons and intrinsic evaluations, we aim to clarify performance and knowledge-transfer differences across different training regimes in comparable conditions. An explicit characterisation of the conditions of bilingual pretraining in models like B-GPT–like models could guide merging strategies that avoid the costs of pretraining, potentially allowing researchers can explore a wider range of language combinations without incurring expensive pretraining runs. These insights can support improved capabilities in "long-tailed" or data-constrained evaluation contexts, such as very low-resource languages, expanding the scope and accessibility of bilingual modelling research.

2 Model Merging

2.1 The Goldfish (Chang et al., 2024) and Bilingual GPT (Arnett et al., 2025) Models

Comparable pretraining frameworks are needed that allow for evaluation across diverse languages and training conditions, due to the heterogeneous architectures and pretraining corpora used in bilingual pretraining and large-scale model merging. As surveyed in Appendix A, pretrained bilingual language models can range from 100M to 130B parameters with diverse training data from learner-like/human-scale data to web-scale sources. We focus on two complementary model families: Goldfish[1] (Chang et al., 2024), a suite of over 1,000 monolingual models covering 350 languages with carefully scaled dataset sizes, and B-GPT[2] (Arnett et al., 2025), a set of bilingual models trained to simulate human-like sequential and simultaneous exposure to L1 and L2.

Goldfish and B-GPT share a systematic pretraining framework based on autoregressive GPT-2–style Transformer architectures, but they differ primarily in the multilingual and bilingual focus. Both use monolingual Unigram tokenizers trained with the SentencePiece library (Kudo and Richardson, 2018) and train on sequences of fixed length with comparable hyperparameters (around 125M parameters, vocab size 50K, max sequence length 128 and512 tokens, respectively). Goldfish comprises over 1,000 monolingual models for 350 languages, trained on dataset sizes scaled by byte premium (Arnett et al., 2024) to ensure comparability (5MB, 10MB, 100MB, 1GB), with multiple epochs to maximise coverage in low-resource languages while maintaining consistent evaluation on FLORES-200. B-GPT, in contrast, targets bilingual pretraining, simulating human-like exposure in bilingual language acquisition via sequential or simultaneous training on deduplicated OSCAR corpora (Suárez et al., 2019) for English paired with Dutch, Spanish, Polish, or Greek, with 2B tokens per language and fine-grained checkpointing to track surprisal dynamics. In bilingual pretraining or merging, two main strategies are applied: sequential merging, where a model is trained on language L1 and then continued on L2 with intermediate checkpoints saved, and simultaneous merging, where models are trained on L1 and L2 separately and later merged into a single bilingual model. By aligning architecture, tokenisation, and evaluation protocols, these models provide a controlled comparison between *merged bilinguality* and *pretrained bilinguality*.

2.2 A School of Bilingual Merged Goldfish Models

There is limited research on creating bilingual models in data-constrained settings. We explore *post hoc* construction of bilingual models by merging independently trained monolingual models with identical architectures, using determinis-

[1]The **Goldfish Models** are available here: https://huggingface.co/goldfish-models

[2]The **BGPT Models** are available here: https://huggingface.co/collections/catherinearnett/b-gpt

tic parameter-space operations (e.g., uniform or weighted averaging). This builds on prior work showing that weight averaging can improve robustness and generalisation without increasing inference cost (Wortsman et al., 2022), and that linear combinations of language- or task-specific models can enable cross-lingual transfer (Bandarkar and Peng, 2025). Ensuring consistency with B-GPT (Arnett et al., 2025), the merged bilingual models are similarly trained using the same architecture and data regime as the GOLDFISH suite of small monolingual models covering 350 languages (Chang et al., 2024) to enables systematic, within-scale comparisons of pretrained bilinguality and merged bilinguality against monolingual baselines. We compare merged bilingual models to joint bilingual pretraining, focusing on merging an English Goldfish model with Spanish, Dutch, Polish, and Greek 1000MB Goldfish models. Simple concatenation or layer averaging often leads to high perplexity and poor cross-lingual generation due to:

1. **Embedding Misalignment**: Monolingual embeddings reside in separate latent spaces, leading to high perplexity when naïvely concatenated.

2. **Transformer Incompatibility**: Coarse-grained layer interpolation can collapse semantic knowledge, particularly in small models.

3. **Tokenizer Differences**: Subword tokenization discrepancies across languages can produce meaningless embeddings if not reconciled.

Our comparison deliberately stresses the limits of current merging techniques.

- **Small Capacity:** The 125M parameter count provides limited redundancy, maximizing the risk of catastrophic parameter interference.

- **Low Data Footprint:** The minimal (5MB–1GB) training data means the models are inherently brittle, making them highly susceptible to forgetting under parameter averaging or interpolation.

Naïve layer interpolation conflates language-specific syntax and semantics, while misaligned embeddings increase perplexity and reduce cross-lingual coherence. We introduce a method that systematically aligns embeddings and merges layers while preserving monolingual structure. First, we do ***embedding alignment*** to identify shared vocabulary tokens across languages, defined as tokens occurring in both monolingual vocabularies. A Procrustes rotation aligns the second language embedding space to the first. High-frequency tokens are augmented with contextual prototypes derived from averaged hidden states; rare tokens are represented by mean subword embeddings after alignment. We then apply ***Transformer Layer Merging.*** We experiment systematically on layer-wise merging strategies to investigate empirical benefits of full merging compared to late merging, which might allow lower layers to preserve monolingual syntactic knowledge, and upper layers blend semantic representations. Specifically, the interpolation weight $\alpha(l, L)$ for layer l of L total layers follows

$$\alpha(l, L) = \alpha_{\text{start}} \cdot \left(1 - \frac{l}{L-1}\right) + \alpha_{\text{end}} \cdot \frac{l}{L-1},$$

where $\alpha_{\text{start}} = 0.7$ and $\alpha_{\text{end}} = 0.3$. Attention, MLP, and layer normalization weights are interpolated separately to ensure numerical stability. Finally, we optionally apply ***Parameter-Efficient Fine-Tuning.***, using LoRA adapters (Hu et al., 2021) with rank $r = 16$ and $\alpha = 32$ are attached to attention and MLP modules. Lower layers (60%) are frozen, and top layers plus LoRA adapters are fine-tuned on a small bilingual corpus of 50k aligned sentence pairs from OPUS (Tiedemann, 2012; Zhang et al., 2020). This is formalised in Algorithm 1, and compared to other merging strategies in *Appendix* B.

3 Evaluation

We evaluate monolingual Goldfish models, bilingual GPT models (bilingual pretrained models) and merged Goldfish models on intrinsic linguistic evaluation benchmarks. Following Jumelet et al. (2025a), we evaluate bilingual and monolingual models on tasks that target formal and functional linguistic competence. Formal competence benchmarks include *mono-BLiMPs*, or language-specific syntactic minimal pairs sets (English BLiMP (Warstadt et al., 2020), BLiMP-NL (Suijkerbuijk et al., 2025)), and MultiBLiMP 1.0 (Jumelet et al., 2025b). MultiBLiMP is a massively multilingual minimal pairs dataset that targets subject-verb agreement for 101 languages,

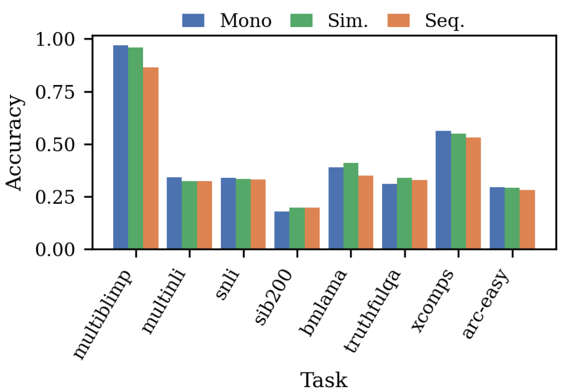

Figure 1: Average accuracy across multilingual and reasoning benchmarks for three training regimes. We compare a monolingual Goldfish baseline with simultaneous and sequential bilingual GPT (B-GPT) models across eight evaluation tasks: Multilingual BLiMP (multiblimp), natural language inference (MultiNLI, SNLI), cross-lingual semantic similarity (SIB-200), factual and relational knowledge probing (BMLAMA), truthfulness assessment (TruthfulQA), compositional generalization (XComps), and elementary reasoning (ARC-Easy). Bars show mean accuracy, averaged over multiple checkpoints or evaluation runs per model and task.

automatically generated from the Universal Dependencies treebanks (Nivre et al., 2017). Functional tasks like BMLAMA (Qi et al., 2023) evaluate factual and domain-specific knowledge memorized by the model, while other functional tasks such as XNLI (Conneau et al., 2018), MultiNLI (Williams et al., 2018), TruthfulQA (Lin et al., 2022), SIB-200 (Adelani et al., 2023) and ARC (Clark et al., 2018) evaluate general reasoning abilities, including natural language inference, commonsense reasoning, narrative understanding, and reading comprehension. Additionally, we evaluate models on XCOMPS (He et al., 2025), a multilingual conceptual minimal pair dataset with 17 languages.

3.1 Effects of Bilingual Pretraining on BLiMP and Cross-Lingual Performance

We evaluated sequential and simultaneous B-GPT models across multiple benchmarks, including MultiBLiMP, BLiMP, and BMLAMA, to assess the impact of bilingual training strategy on task performance and cross-lingual retention (Tables 1, 5, 9). Sequential B-GPT exhibits catastrophic forgetting on L1, particularly when English is the L1, and shows variable performance on reasoning and semantic tasks. In contrast, simultaneous B-GPT maintains more consistent accuracy across tasks, closer to monolingual baselines, and mitigates interference.

To test whether bilingual models systematically outperform or underperform monolingual models, we compared B-GPT models to the monolingual Goldfish baseline across eight multilingual and reasoning benchmarks. For each language, we computed score differences and summarized them with the mean, standard deviation, and Cohen's d, applying one-sided t-tests (p_{up}, p_{down}) for directional hypotheses (H1: bilingual > monolingual; H2: bilingual < monolingual). Multiple comparisons were corrected using Bonferroni and Benjamini-Hochberg FDR procedures ($\alpha = 0.05$). After correction, no language showed a statistically significant bilingual advantage. Modest positive trends were observed for Polish ($\Delta\bar{x} = 0.018$, $d \approx 0.43$, $p = 0.064$) and Greek ($\Delta\bar{x} \approx 0.006$), while English and Spanish showed negligible negative differences ($\Delta\bar{x}_{\text{EN}} \approx -0.0002$, $\Delta\bar{x}_{\text{ES}} \approx -0.008$).

BLiMP-specific analyses further support these patterns. A linear mixed-effects model with BLiMP phenomenon as a random intercept was fitted, including fixed effects for training type (sequential vs. simultaneous), English status (L1 vs. L2), non-English language (Dutch, Greek, Polish, Spanish), and their two-way interactions. The intercept was significantly above zero ($\beta = 0.65$, $SE = 0.03$, $t = 26.1$, $p < 0.001$), indicating sequential B-GPT EN–NL models perform above chance. Non-English language had no significant effect. When English was the L1, simultaneous training significantly increased accuracy relative to sequential ($\beta = 0.15$, $SE = 0.01$, $t = 16.4$, $p < 0.001$), whereas when English was the L2, sequential training yielded higher accuracy than simultaneous ($\beta = -0.16$, $SE = 0.01$, $t = -19.2$, $p < 0.001$). These effects were consistent across non-English languages (Tables 5, 9; Figure 2).

Across nearly all BLiMP phenomena in Dutch and English, monolingual models outperformed bilingual models, with sequential B-GPT showing particularly low accuracy on determiners and topicalization, reflecting catastrophic interference. Simultaneous B-GPT partially mitigated these deficits, improving performance on auxiliaries, infinitival clauses, and passive constructions, but generally did not reach monolingual ceilings (Table 9).

Evaluation on BLiMP-NL confirmed these patterns. A Friedman test revealed significant overall differences across five models ($X^2(4) = 64.94$, $p < .001$). Pairwise Wilcoxon signed-rank tests with Holm-Bonferroni correction showed that the monolingual Dutch model significantly outperformed all B-GPT models ($p < .001$), except for the NL-EN simultaneous model ($p = 1$). Among Dutch bilingual models, training type did not significantly affect accuracy when Dutch was the L2 (EN–NL sequential vs. EN–NL simultaneous; $p = .452$), but when Dutch was the L1, simultaneous training significantly improved accuracy ($p < .001$), mirroring the English BLiMP pattern.

These results indicate that while bilingual exposure alone does not guarantee a systematic advantage, simultaneous training reduces L1 interference and supports more robust cross-lingual generalisation under fixed model capacity and evaluation conditions.

	language(s)	multiblimp (EN)	multiblimp (Other)	BLiMP (EN)
Chance		0.5	0.5	0.5
Monolingual Goldfish	EN	0.964	-	0.78
	NL	0.701	0.973	-
	ES	0.656	0.961	-
	EL	0.640	0.988	-
	PL	0.623	0.963	-
Sequential B-GPT	EN-NL	0.8416	0.9725	0.64
	EN-ES	0.7766	0.9516	0.632
	EN-EL	0.8494	0.9845	0.66
	EN-PL	0.8052	0.9471	0.638
	NL-EN	0.9662	0.7224	0.794
	ES-EN	0.9662	0.7855	0.789
	EL-EN	0.965	0.666	0.776
	PL-EN	0.9636	0.6537	0.777
Simultaneous B-GPT	EN-NL	0.969	0.961	0.795
	EN-ES	0.97	0.937	0.792
	EN-EL	0.969	0.973	0.787
	EN-PL	0.974	0.912	0.779
	NL-EN	0.948	0.976	0.784
	ES-EN	0.955	0.959	0.776
	EL-EN	0.984	0.943	0.757
	PL-EN	0.952	0.956	0.763
Merged	EN-NL	0.652	0.570	0.514
	EN-ES	0.626	0.668	-
	EN-EL	0.608	0.621	0.5289
	EN-PL	0.657	0.586	0.5197

Table 1: Performance of the monolingual Goldfish models (Chang et al., 2024) and B-GPT models (Arnett et al., 2024) across MultiBLiMP 1.0 (Jumelet et al., 2025b) and English BLiMP benchmarks (Warstadt et al., 2020). "Chance" indicates random baseline performance. Monolingual Goldfish models achieve near-ceiling accuracy on their respective languages, while Sequential and Simultaneous B-GPT models show varying degrees of cross-lingual transfer. The merged model exhibits lower performance overall.

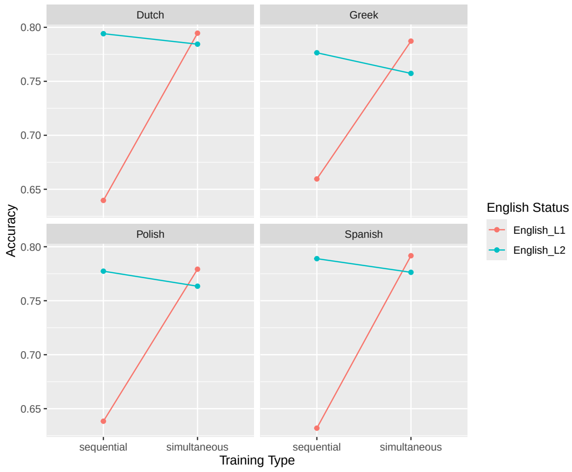

Figure 2: Average accuracy on BLiMP dataset for B-GPT models across training type conditions and English status.Sequential B-GPT models exhibit lower accuracy when English is the L1, reflecting catastrophic forgetting, whereas simultaneous B-GPT models maintain higher and more consistent performance across both L1 and L2 conditions. When English is the L2, sequential training leads to higher accuracy than simultaneous training, as indicated by the negative interaction in the mixed-effects model (Table 5). These patterns hold across multiple non-English languages (Greek, Polish, Spanish), suggesting that simultaneous bilingual training mitigates interference and supports better cross-lingual generalisation, while sequential training results in L1 retention deficits and variable task performance.

3.2 Merged Model Performance

Additionally, the merged models were tested on BLiMP (Table 1). Accuracy was compared to the monolingual English model and the B-GPT models. A Friedman test revealed a significant overall difference between models ($X^2(20) = 657.77$, $p<.001$). Pairwise Wilcoxon signed-rank tests (Holm-Bonferroni p-correction) were performed to compare accuracy of the merged model to the monolingual English model and its respective B-GPT models (e.g., EN-NL merged model is compared to monolingual English, EN-NL sequential B-GPT, EN-NL simultaneous B-GPT, NL-EN sequential B-GPT, and NL-EN simultaneous B-GPT). BLiMP accuracy for merged models was significantly different to the monolingual model and all its respective B-GPT models for all language pairs ($p <.001$). The merged models performed significantly *worse* than the monolingual and B-GPT models. Comparing merged performance of non-English monolingual Goldfish models and merged models, the

Language Pair	L1/L2	LoRA	multiblimp	xnli	mnli_local	mnli_en	snli_en	snli_local	sib200	bmlama	truthfulqa	xcomps	arc-easy
en_pl	L1 (en)	+	0.658	0.335	0.358	0.358	0.361	0.361	0.167	0.194	0.235		0.259
	L2 (pl)	+	0.585		0.319	0.322	0.357	0.357	0.172	0.162	0.207		0.251
	L1 (en)	-	0.657	0.334	0.358	0.358	0.361	0.361	0.167	0.197	0.235		0.260
	L2 (pl)	-	0.586		0.320	0.321	0.357	0.357	0.172	0.163	0.206		0.251
en_nl	L1 (en)	+	0.653	0.339	0.358	0.358	0.362	0.362	0.191	0.162	0.223		0.260
	L2 (nl)	+	0.573		0.359	0.359	0.357	0.357	0.196	0.170	0.224	0.495	0.258
	L1 (en)	-	0.652	0.338	0.358	0.358	0.364	0.364	0.191	0.164	0.224		0.260
	L2 (nl)	-	0.570		0.359	0.359	0.357	0.357	0.201	0.170	0.224	0.495	0.256
en_el	L1 (en)	+	0.606	0.329	0.346	0.346	0.324	0.324	0.127	0.211	0.224		0.261
	L2 (el)	+	0.622	0.337	0.321	0.341	0.323	0.329	0.152	0.198	0.243	0.486	0.245
	L1 (en)	-	0.608	0.333	0.347	0.347	0.326	0.326	0.127	0.211	0.224		0.262
	L2 (el)	-	0.621	0.336	0.321	0.341	0.324	0.329	0.152	0.197	0.238	0.485	0.244
en_es	L1 (en)	+	0.625	0.335	0.326	0.326	0.320	0.320	0.118	0.123	0.216		0.259
	L2 (es)	+	0.672	0.336	0.326	0.327	0.315	0.315	0.137	0.120	0.226	0.514	0.272
	L1 (en)	-	0.626	0.335	0.325	0.325	0.318	0.318	0.118	0.125	0.218		0.261
	L2 (es)	-	0.668	0.336	0.327	0.327	0.316	0.315	0.137	0.121	0.224	0.515	0.272

Table 2: Evaluation of Model Merging (Section 2.2) across , +LoRA (merge-trained) or -LoRA (merge-merged) on *multiblimp* (Warstadt et al., 2020; Suijkerbuijk et al., 2025; Jumelet et al., 2025b), *xnli* (Conneau et al., 2018), *mnli_local / mnli_en* (Williams et al., 2018), *snli_en / snli_local* (Williams et al., 2018), *sib200* (Adelani et al., 2023), *bmlama* (Qi et al., 2023), *truthfulqa* (Lin et al., 2022), *xcomps* (He et al., 2025), *arc-easy* (Clark et al., 2018).

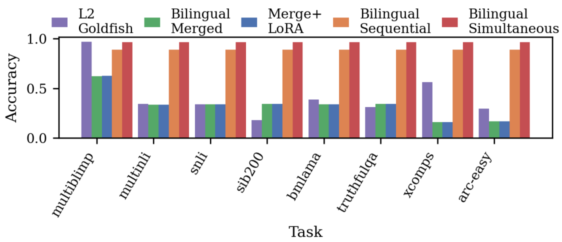

Figure 3: Average task performance across Goldfish (non-English, L2), BGPT sequential vs. simultaneous training and bilingual merged models (± LoRA). Bars show mean accuracy per task over languages.

merged models show only minor changes in performance—slightly lower for NL (0.570 vs 0.701), EL (0.621 vs 0.640), and PL (0.586 vs 0.623) and marginally higher for ES (0.668 vs 0.656)—indicating that naive weight merging does not substantially improve non-English capabilities beyond weak monolingual performance. Full performance is compared in *Figure* 3) and *Table* 2.

4 Discussion and Conclusion

4.1 The Curse of Bilinguality?

Our findings provide a systematic investigation of bilingual pretraining and model merging in small-capacity, data-constrained settings, directly addressing the research gaps outlined in the Introduction. Consistent with the literature on joint multilingual pretraining (e.g., Conneau et al., 2020a), simultaneous bilingual training supports the creation of a shared representational space that enables cross-lingual transfer and mitigates catastrophic forgetting in the L1. In contrast, sequential bilingual training often results in negative interference, consistent with the “curse of bilinguality” (Zhou and Matusevych, 2025), where adding a second language can degrade monolingual performance. Simultaneous bilingual training consistently benefits L1 performance across languages, while sequential training can sometimes improve L2 at the cost of L1, with effects modulated by typology, script, and tokenization. These results support the notion that training regime—not just bilingual exposure—crucially shapes cross-lingual transfer, with simultaneous pretraining preserving L1 competence and mitigating interference, whereas sequential pretraining risks negative transfer, especially for English as L1.

4.2 Bilingual Model Merging

Incremental bilingual adaptation via model merging offers a practical trade-off, while sacrificing some theoretical optimality for substantial real-world benefits: lower compute and data requirements, faster iteration, and preservation of the base model's monolingual strengths. However, naive merging as such a small-scale illustrates the challenges of this approach in data-constrained regimes, as it can degrade monolingual performance and fail to capture bilingual nuances, particularly for rare or low-frequency constructions. This underscores the need for systematic, small-scale merging experiments—carefully calibrating merge weights, using high-quality anchor lexica, and monitoring both monolingual and bilingual metrics—where sophisticated strategies like Fisher-weighted merging or layer-specific methods such as RegMean can guide parameter selection and gradually build robust bilingual models. At larger scales, scratch pretraining remains valuable, providing a globally consistent bilingual lexicon and fully balanced inductive bias, but incremental adaptation remains the more efficient and cost-effective solution when compute, data, or production stability are limiting factors.

4.3 Conclusion

This work underscores the importance of designing benchmarks that systematically compare joint pretraining, sequential pretraining, and model merging under identical, data-constrained conditions, accounting for typological diversity and avoiding over-representation of high-resource languages. Our experiments demonstrate that small-capacity bilingual models are highly sensitive to training regime and integration strategy: simultaneous pretraining preserves L1 while enabling cross-lingual transfer, whereas sequential pretraining or naive model merging can induce negative interference. These findings highlight how incremental model merging can be rigorously evaluated in low-data, small-capacity settings without compromising monolingual performance, and demonstrate the need for fine-grained benchmarking to capture subtle differences between small-scale bilingual pretraining and merged models. Small-capacity models require careful, controlled evaluation to fairly assess both successful transfer and negative interference, particularly in low-resource languages. Overall, this framework supports systematic small-scale experiments to quantify trade-offs between efficiency, cross-lingual transfer, and L1 preservation, informing both model development and multilingual evaluation practices.

5 Limitations

Despite the insights gained from our study of data-constrained bilingual language models, several limitations should be noted. First, vocabulary and tokenization misalignment remains a challenge. Even with Procrustes alignment, differences in subword tokenization across languages can leave some embeddings poorly aligned. Rare or language-specific subwords may lack adequate anchors, limiting the quality of cross-lingual transfer and increasing perplexity in merged models.

Second, our evaluation is constrained by limited coverage of linguistic phenomena. While benchmarks such as BLiMP, MultiBLiMP, and BM-LAMA capture key aspects of formal and functional competence, they do not exhaustively assess low-frequency constructions, idiomatic expressions, or morphologically rich languages. Consequently, model performance in these untested areas remains uncertain. We suggest that language-specific formal and functional competence benchmarks (e.g., monolingual BLiMP-style evaluations, like BLiMP-NL) are needed for a more comprehensive and interpretable evaluation.

Third, the sensitivity to merge hyperparameters is notable. Layer-wise interpolation weights and LoRA adaptation ranks influence merged model performance. Small deviations can substantially affect L1 preservation and L2 transfer, indicating that naive merging is brittle and requires careful hyperparameter tuning for each language pair. Additionally, as demonstrated in our experiments, rare words and low-frequency tokens remain underrepresented. Small bilingual corpora exacerbate this effect, biasing merged models toward frequent patterns and limiting coverage of less common linguistic constructions.

Finally, small model capacity and naive merging strategies pose further limitations. With only 125M parameters, merged models have limited redundancy and are prone to interference, constraining the effectiveness of uniform parameter averaging. Naive merging can collapse semantic knowledge or misalign embeddings, suggesting that more principled strategies—such as Fisher-weighted merging, RegMean, or layer-wise selective approaches—could improve performance, but were not fully explored in this study.

Collectively, these limitations highlight the challenges of bilingual model merging in low-data, small-capacity settings and underscore the need for careful design choices, hyperparameter tuning, and future exploration of more sophisticated merging strategies.

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A Related Work: Bilingual Pretraining and Model Merging

A.1 Bilingual Pretraining

Pretrained Bilingual Language Models – such as CEDILLE [3] (French-English; Müller and Laurent (2022)), CROISSANTLLM (French-English; Faysse et al. (2024)) and SABIÁ [4] (Portuguese-English Pires et al. (2023)) – are explicitly designed to handle only two languages to reduce the need for parallel corpora, or to handle code-mixed input. Bilingual pretrained models range in model size. Large-scale Chinese–English pretrained models **GLM-130B**[5] (Zeng et al., 2022) and **MAP-Neo**[6] (Zhang et al., 2024) serving high-resource bilingual scenarios. Other efforts including FLOR (Spanish/Catalan-English; Da Dalt et al. (2024), LLAMAS (English-Estonian; Kuulmets et al. (2024) and ROQLLAMA (Romanian-English; Dima et al. (2024)) use bilingual models to support lower-resourced languages. Bilingual pretrained models have been developed as cognitively-inspired models of L2 acquisition, or L2LMs, as computational models of bilingual or second language acquisition, often in similar human-scale pretraining constraints to the BabyLM Challenge (Gao et al., 2025). These investigate questions of transfer (Yadavalli et al., 2023; Oba et al., 2023), or investigate the effects of cognitive priors (e.g., alignment to learner reading times; preserving L1 knowledge to probe the Critical Period Hypothesis) and compare sequential vs. mixed L1/L2 exposure (Aoyama and Schneider, 2024; Constantinescu et al.; Arnett et al., 2025).

A.2 Model Merging in Multilingual NLP

Model merging – the practice of combining weights from independently trained models into a single unified model – has gained traction as a compelling alternative in multilingual NLP to traditional ensembling and multitask training. This builds on empirical findings that averaging weights of multiple fine-tuned models can improve robustness and generalisation without incurring extra inference costs (Wortsman et al., 2022). This has recently motivated weight space fusion techniques: Bandarkar and Peng (2025) suggest that post-hoc model merging and modular merging strategies can be "surprisingly effective" for cross-lingual transfer in large language models, indicating that linear combinations of expert models trained on different languages or tasks can capture complementary capabilities, alongside other attempts to adapt multilingual models to code-mixed tasks via merging weights (Kodali et al., 2025).

B Model Merging Methods

B.1 Method 1

The first merging method, *layer-wise merging with embedding alignment and LoRA fine-tuning*, efficiently combines monolingual models into a stable bilingual language model. This approach integrates Procrustes-aligned embeddings, depth-aware layer interpolation, and parameter-efficient adaptation, while minimizing computational overhead through embedding alignment, contextual prototypes, and selective fine-tuning. The full pipeline is formalised in Algorithm 1.

Embedding Alignment Shared tokens across languages are aligned via a Procrustes rotation to reconcile embedding spaces. High-frequency tokens are augmented with *contextual prototypes*, computed as averaged hidden states from monolingual models to capture usage patterns beyond static embeddings. Rare tokens are represented by the mean of subword embeddings after alignment. Formally, for the Spanish embedding matrix E_2, we compute:

$$E_2^{\text{aligned}} = R \cdot E_2,$$

[3] https://huggingface.co/Cedille/fr-boris

[4] https://huggingface.co/maritaca-ai/sabia-7b

[5] https://github.com/THUDM/GLM-130B

[6] https://github.com/multimodal-art-projection/MAP-NEO

where R is the Procrustes rotation mapping shared vocabulary to the English embedding space E_1. Contextual prototypes for token t are computed as

$$\mathbf{c}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} h_i^{(t)},$$

where $h_i^{(t)}$ are hidden states of t across N_t example contexts from monolingual models.

Transformer Layer Merging Transformer layers are merged using a depth-aware interpolation:

$$\text{merged\_layer}_\ell = \alpha_\ell \cdot \text{L1\_layer}_\ell + (1 - \alpha_\ell) \cdot \text{L2\_layer}_\ell,$$

where α_ℓ decays from lower to upper layers. Lower layers preserve syntactic and structural features, while upper layers blend semantic content. Attention, MLP, and layer normalization weights are interpolated separately to maintain numerical stability. Optionally, only the top fraction of layers can be merged, controlled by a `merge_top_frac` hyperparameter.

Parameter-Efficient Fine-Tuning LoRA adapters are applied to key modules (attention projections and MLP layers), while lower transformer layers (typically $\sim 60\%$) are frozen to preserve monolingual knowledge. Upper layers and LoRA parameters are fine-tuned on a small bilingual corpus (e.g., 50k sentence pairs from OPUS Books) to consolidate shared and language-specific representations. Text is tokenized using a pretrained bilingual tokenizer to ensure compatibility of embeddings and the LM head. Learning rate, batch size, gradient accumulation, and LoRA hyperparameters are configurable.

Top-Layer Fraction Merging Let L be the total number of transformer layers and `merge_top_frac` $\in [0, 1]$ the fraction of top layers to merge. Then, the merge set is:

$$\mathcal{L}_{\text{merge}} = \Big\{ \ell \mid \ell \geq L \cdot (1 - \texttt{merge\_top\_frac}) \Big\}.$$

Interpolation Coefficient per Layer For each layer $\ell \in \mathcal{L}_{\text{merge}}$, the interpolation coefficient α_ℓ is computed linearly between a starting weight α_{start} and ending weight α_{end}:

$$\alpha_\ell = \alpha_{\text{start}} \cdot \left(1 - \frac{\ell - \ell_{\min}}{\ell_{\max} - \ell_{\min}}\right) + \alpha_{\text{end}} \cdot \frac{\ell - \ell_{\min}}{\ell_{\max} - \ell_{\min}},$$

where $\ell_{\min}$ and $\ell_{\max}$ are the minimum and maximum indices of layers being merged. This ensures lower layers are weighted toward L1 (preserving structural features) and higher layers progressively incorporate L2 features.

Merged Layer Computation For attention (A_ℓ), MLP (M_ℓ), and optionally layer norm parameters, the merged layer is:

$$\begin{aligned}
A_\ell^{\text{merged}} &= \alpha_\ell \, A_\ell^{\text{L1}} + (1 - \alpha_\ell) \, A_\ell^{\text{L2}}, \\
M_\ell^{\text{merged}} &= \alpha_\ell \, M_\ell^{\text{L1}} + (1 - \alpha_\ell) \, M_\ell^{\text{L2}}, \\
\text{LN}_\ell^{\text{merged}} &= \alpha_\ell \, \text{LN}_\ell^{\text{L1}} + (1 - \alpha_\ell) \, \text{LN}_\ell^{\text{L2}}.
\end{aligned}$$

Hyperparameter Sweep To systematically explore the effect of merging hyperparameters on bilingual model quality, we perform a grid search over key factors, including start/end layer interpolation weights, LoRA rank and alpha, and layer freeze ratio. Each checkpoint can be optionally pushed to the Hugging Face Hub. This systematic hyperparameter search allows us to identify optimal combinations of start/end weights, LoRA configuration, and freeze ratios for bilingual merging. By controlling the fraction of top layers merged and the layer-wise interpolation coefficients, we maintain structural features from L1 in lower layers while blending semantic features from L2 in upper layers. All merged checkpoints and metrics are tracked in a CSV for downstream analysis and Hugging Face sharing.

Algorithm 1 Bilingual Model Merging + LoRA Fine-Tuning with Top-Layer Fraction Merging

Require: Monolingual models M_1 (L1), M_2 (L2), tokenizers TOK_1, TOK_2; bilingual tokenizer TOK_{bi}; bilingual corpus $\mathcal{D}_{\text{bi}}$; hyperparameters $\alpha_{\text{start}}, \alpha_{\text{end}}, f_{\text{top}}, r_{\text{LoRA}}, \alpha_{\text{LoRA}}$, freeze_ratio

$V_{\text{bi}} \leftarrow \text{VOCAB}(TOK_{\text{bi}}), T_s \leftarrow TOK_1 \cap TOK_2$

Extract embeddings E_1, E_2; compute Procrustes R on shared tokens T_s: $E_2[T_s]R \approx E_1[T_s]$; $E_2 \leftarrow RE_2$

for $t \in V_{\text{bi}}$ **do**

  $e_1 \leftarrow E_1[t],\ e_2 \leftarrow E_2[t]$

  $E_{\text{merged}}[t] \leftarrow \begin{cases} 0.5(e_1 + c_1) + 0.5(e_2 + c_2), & t \text{ high-frequency shared} \\ 0.5(e_1 + e_2), & \text{otherwise} \end{cases}$

end for

Initialize merged model M_{merged} with E_{merged} for input embeddings and LM head

$L \leftarrow$ number of transformer layers in M_{merged}

$\ell_{\min} \leftarrow \lfloor L \cdot (1 - f_{\text{top}}) \rfloor, \quad \ell_{\max} \leftarrow L - 1$

for $\ell = \ell_{\min} \dots \ell_{\max}$ **do**

  Compute layer-wise interpolation coefficient:

$$\alpha_\ell = \alpha_{\text{start}} \cdot \left(1 - \frac{\ell - \ell_{\min}}{\ell_{\max} - \ell_{\min}}\right) + \alpha_{\text{end}} \cdot \frac{\ell - \ell_{\min}}{\ell_{\max} - \ell_{\min}}$$

  Merge weights:

$$A_\ell \leftarrow \alpha_\ell A_\ell^1 + (1 - \alpha_\ell) A_\ell^2, \quad M_\ell \leftarrow \alpha_\ell M_\ell^1 + (1 - \alpha_\ell) M_\ell^2$$

  Merge layer normalization parameters similarly

end for

Attach LoRA adapters to {c_attn, c_proj, mlp.c_fc, mlp.c_proj} with rank r_{LoRA} and α_{LoRA}

Freeze lower $\lfloor L \cdot \text{freeze\_ratio} \rfloor$ layers

Fine-tune on $\mathcal{D}_{\text{bi}}$: train LoRA adapters + top layers

Evaluate model: perplexity, fragmentation, entropy, cross-lingual generation

return Fine-tuned bilingual model

B.2 Method 2: Tokeniser-Aware Attention-Head Selective Merging

Motivation. Uniform layer interpolation treats all transformer subcomponents—embeddings, attention heads, and MLPs—as equally mergeable across monolingual models. This coarse-grained averaging often fails to respect language-specific patterns in lower layers and tokenization differences, which can lead to semantic collapse despite preserving surface fluency. Directly averaging embeddings without accounting for subword tokenizers like SentencePiece can produce nonsensical vectors, especially for ambiguous or code-switched tokens.

Procedure. Method 2 selectively merges embeddings and attention heads to preserve both shared and language-specific knowledge. We propose a tokenizer-aware, head-similarity guided merge that selectively combines model components based on data-driven criteria. Each token is encoded under both monolingual tokenizers, and embeddings are merged in vector space rather than by index, allowing the model to handle ambiguous or code-switched tokens via self-attention. A tokenwise interpolation weight α_{L1} (swept between 0.1–0.9) balances contributions from L1 and L2 embeddings. Attention matrices are then compared using cosine similarity on a small bilingual corpus, and only heads exceeding a threshold ($\tau = 0.85$) are interpolated; other heads retain their monolingual weights, preserving language-specific knowledge. As in Method 1, LoRA adapters (rank $r = 16$, $\alpha = 32$) are optionally applied. Method 2 is formalised in Algorithm 2.

Method. Let M_1 and M_2 be monolingual models with tokenizers T_1 and T_2, and let $\mathcal{D}_{bi}$ be a small bilingual corpus. The merged model M_{bi} is constructed as follows:

Algorithm 2 Tokenizer-Aware, Head-Similarity Guided Merge

Require: Monolingual models M_1, M_2; tokenizers T_1, T_2; bilingual corpus $\mathcal{D}_{bi}$; merge weight α; similarity threshold τ
Ensure: Merged bilingual model M_{bi}
Step 1: Tokenizer-Aware Embedding Merge
for token t in shared tokenizer T_{bi} **do**
    $v_1 \leftarrow \text{mean}(\text{embeddings of } T_1(t) \text{ in } M_1)$
    $v_2 \leftarrow \text{mean}(\text{embeddings of } T_2(t) \text{ in } M_2)$
    $v_{bi}[t] \leftarrow \alpha \cdot v_1 + (1 - \alpha) \cdot v_2$
end for
Inject v_{bi} as the embedding matrix in M_{bi}; optionally tie LM head
Step 2: Attention Head Similarity Analysis
for layer $l = 1 \dots L$ and head $h = 1 \dots H$ **do**
    Compute attention matrices A_1, A_2 on $\mathcal{D}_{bi}$ for head h in layer l
    Compute similarity $s_{l,h} = \text{cosine\_similarity}(A_1, A_2)$
end for
Step 3: Selective Head Merge
for layer $l = 1 \dots L$ and head $h = 1 \dots H$ **do**
    if $s_{l,h} \geq \tau$ **then**
        Interpolate head weights: $W_{bi}^{(l,h)} = \alpha W_1^{(l,h)} + (1 - \alpha) W_2^{(l,h)}$
    else
        Keep original weights from M_1 (or M_2)
    end if
end for
Step 4: Fine-Tuning
Apply LoRA to selected modules and optionally freeze lower layers
Fine-tune M_{bi} on $\mathcal{D}_{bi}$ for bilingual adaptation
return M_{bi}

Performance on MultiBLiMP is shown in *Table* 3.

Model	L2	English	L2 Score
`suchirsalhan/en_pl-alpha10-TOKENWISE_MERGE`	pl	0.639	0.598
`suchirsalhan/en_nl-alpha10-TOKENWISE_MERGE`	nl	0.636	0.568
`suchirsalhan/en_el-alpha10-TOKENWISE_MERGE`	el	0.588	0.637
`suchirsalhan/en_es-alpha10-TOKENWISE_MERGE`	es	0.631	0.684

Table 3: MultiBLiMP scores for English and L2 in bilingual merged models.

C Statistical Significance

C.1 Per Task

Table 4: Meta-analytic t-tests across metrics per language (with multiple comparisons correction). Positive mean differences indicate bilingual models outperforming monolingual models.

Language	Mean Diff	Cohen's d	p_{up}	$p_{\mathrm{up}}^{\mathrm{Bonf}}$	$p_{\mathrm{up}}^{\mathrm{FDR}}$	$\mathrm{Sig}_{\mathrm{up}}^{\mathrm{Bonf}}$	$\mathrm{Sig}_{\mathrm{up}}^{\mathrm{FDR}}$	p_{down}	$p_{\mathrm{down}}^{\mathrm{Bonf}}$	$p_{\mathrm{down}}^{\mathrm{FDR}}$	$\mathrm{Sig}_{\mathrm{down}}^{\mathrm{Bonf}}$	$\mathrm{Sig}_{\mathrm{d}}^{\mathrm{F}}$
EN	-0.000157	-0.0062	0.5090	1.0000	0.6787	False	False	0.4910	1.0000	0.9361	False	Fa
ES	-0.007994	-0.2657	0.8477	1.0000	0.8477	False	False	0.1523	0.6093	0.6093	False	Fa
EL	0.005581	0.1711	0.2521	1.0000	0.5042	False	False	0.7479	1.0000	0.9361	False	Fa
PL	0.018186	0.4347	0.0639	0.2556	0.2556	False	False	0.9361	1.0000	0.9361	False	Fa

C.2 Linear Mixed Effects

To assess whether bilingual GPT models systematically outperform or underperform monolingual models, we compared the monolingual Goldfish baseline with sequential and simultaneous B-GPT models across eight multilingual and reasoning benchmarks (see Table in Appendix). For each language, we formulated two directional hypotheses: H1, that bilingual models improve over monolingual performance, and H2, that they perform worse. We computed score differences across all metrics, summarizing these with the mean, standard deviation, and Cohen's d, and applied one-sided t-tests to evaluate support for H1 or H2 (p_{up} and p_{down}). Multiple comparisons were corrected using Bonferroni and Benjamini-Hochberg FDR procedures ($\alpha = 0.05$). After correction, no language showed a statistically significant bilingual advantage. Polish (PL) exhibited the largest positive trend ($\Delta\bar{x} = 0.018$, $d \approx 0.43$, $p = 0.064$), suggesting a modest potential benefit, whereas English (EN) and Spanish (ES) showed negligible negative differences ($\Delta\bar{x}$EN ≈ -0.0002, $\Delta\bar{x}$ES ≈ -0.008), and Greek (EL) displayed a small positive difference ($\Delta\bar{x}_{\mathrm{EL}} \approx 0.006$). These results highlight that, although some trends exist, there is no robust evidence for a consistent bilingual advantage across languages once multiple comparisons are taken into account.Because MultiBLiMP evaluates only subject-verb agreement, we further compared monolingual English performance to all 17 B-GPT bilingual models on BLiMP (Warstadt et al., 2020), covering 67 linguistic phenomena. A Friedman test revealed a significant model effect ($X^2(16) = 403.299$, $p < .001$), followed by pairwise Wilcoxon signed-rank tests with Holm-Bonferroni correction. Sequential B-GPT models matched monolingual performance when English was L2 ($p \geq .051$) but underperformed when English was L1 ($p \leq .001$). Simultaneous B-GPT models generally maintained monolingual-level performance, with exceptions for Greek L2 and Dutch L2. To further capture these patterns, we fitted a linear mixed-effects model with fixed effects for training type (sequential vs. simultaneous), English status (L1 vs. L2), and other language (Dutch, Spanish, Greek, Polish), including BLiMP phenomenon as a random intercept. The best-fit model included two significant two-way interactions: training type × English status and training type × other language. A three-way interaction did not improve model fit ($p = .17$). These results indicate that, although bilingual models do not consistently outperform monolingual models, simultaneous bilingual training supports better cross-lingual retention, robustness, and generalisation across diverse tasks (see Table 5). Since MultiBLiMP is limited in its evaluation of subject-verb agreement, we further evaluate BGPT and Goldfish models on BLiMP-NL (*Table* 9), which is the only language-specific BLiMP beyond English for the pretrained BGPT models. Across nearly all grammatical phenomena, including auxiliaries, adpositional phrases, and infinitival argument clauses, the monolingual model achieves substantially higher accuracy, often approaching ceiling performance, whereas the bilingual models lag behind, even under simultaneous training. While simultaneous B-GPT generally outperforms sequential B-GPT on several phenomena, such as auxiliaries, infinitival clauses, and passive constructions, it still fails to match the monolingual baseline, indicating that bilingual exposure alone does not suffice to fully capture language-specific syntactic knowledge. Certain constructions, like determiners and topicalisation, show particularly low performance for bilingual models, highlighting areas where cross-lingual transfer is limited or where catastrophic interference may occur. To test whether this finding extends to languages beyond English, five models - the Dutch goldfish and the four B-GPT models - were evaluated on the BLiMP-NL dataset (Suijkerbuijk et al., 2025) (see Table **??** in Appendix **X** for results). A Friedman test revealed a significant overall difference between models ($X^2(4)$ = 64.94, p < .001). Pairwise Wilcoxon signed-rank tests (Holm-Bonferroni p-correction) were performed (see Table 12 in Appendix **X**), and revealed that the monolingual model significantly outperformed all B-GPT models (p<.001) except for the NL-EN simultaneous model, where performance was not significantly different from the monolingual model (p = 1). Amongst the Dutch bilingual models, training type condition did not significantly affect accuracy when Dutch was the L2 (EN-NL sequential vs. EN-NL simultaneous; p = .452) (**different from English BLiMP pattern**). When Dutch was the L1, however, simultaneous training significantly improved BLiMP-NL accuracy compared to sequential training (p <.001) (**same as English BLiMP pattern**).

The intercept was significantly greater than zero (β = 0.65, SE = 0.03, t = 26.1, p <.001), indicating that BLiMP accuracy was significantly above chance for the sequential B-GPT EN-NL model. The type of non-English language did not significantly influence BLiMP accuracy. When English was the L1, accuracy significantly increased with simultaneous training compared to sequential (β = 0.15, SE = 0.01, t = 16.4, p <.001). With sequential training, BLiMP accuracy significantly increases when English was the L2 (β = 0.14, SE = 0.01, t = 24.1, p <.001). With simultaneous training, BLiMP accuracy significantly *decreased* when English was the L2 (β = -0.16, SE = 0.01, t = -19.2, p <.001); this was true regardless of the non-English language. For sequential training, BLiMP accuracy improves when English is the L2 relative to L1. For simultaneous training, BLiMP accuracy *decreases* when English is the L2 relative to L1. When English is the L1, simultaneous training significantly improves performance relative to sequential training; when English is the L2, accuracy is higher with sequential training. This is true with all non-English languages, as illustrated in Figure 2.

Fixed effects	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	0.646	0.025	75.24	26.088	<.001
Simultaneous	0.1524	0.009	996.0	16.419	<.001
English_L2	0.1418	0.006	996.0	24.102	<.001
Greek	0.001	0.008	996.0	0.133	.89
Polish	-0.01	0.008	996.0	-1.086	.28
Spanish	-0.01	0.008	996.0	-0.78	.44
Simultaneous x English_L2	-0.1596	0.008	996.0	-19.18	<.001
Simultaneous x Greek	-0.018	0.012	996.0	-1.56	0.12
Simultaneous x Polish	-0.01	0.012	996.0	-0.78	.44
Simultaneous x Spanish	0.001	0.012	996.0	0.08	.94

Table 5: Linear mixed-effects model summary predicting BLiMP accuracy for B-GPT models across training type (sequential vs. simultaneous), English status (L1 vs. L2), and target non-English languages (Greek, Polish, Spanish). The model includes fixed effects for training type, English status, non-English language, and their two-way interactions; BLiMP phenomena were included as a random intercept (not shown). Estimates indicate the effect of each predictor relative to the reference levels (sequential training and English L1). Significant positive estimates for *Simultaneous* (β = 0.152, p < 0.001) and *English*$_L2(\beta = 0.142$, p < 0.001*) show that simultaneous training improves accuracy when English is the L1 and that sequential training yields higher accuracy when English is the L2. The negative interaction Simultaneous x English*$_L2(\beta = -0.160$, $p < 0.001$*) indicates that simultaneous training reduces accuracy when English is the L2. Non-English language effects and interactions are largely non-significant, suggesting that the observed patterns hold consistently across Greek, Polish, and Spanish.*

D Multilingual Evaluation of Goldfish and B-GPT Models

	multiblimp	blimp-NL	multinli	snli	sib200	bmlama	truthfulqa	xcomps	arc-easy
Random chance	0.5	-	0.33	0.33	0.1429	0.1	0.25	0.5	0.25
English goldfish	0.964	-	0.3545	0.3289	0.1961	0.4699	0.3044	-	0.3044
Dutch goldfish	0.973	0.8773	0.359	0.3567	0.1961	0.4267	0.3112	0.5724	0.2938
Spanish goldfish	0.961	-	0.359	0.3567	0.2255	0.39	0.2857	0.5691	0.2891
Greek goldfish	0.988	-	0.3167	0.3289	0.1471	0.3334	0.3231	0.5413	0.2942
Polish goldfish	0.963	-	0.3167	0.3289	0.1225	0.3172	0.3248	-	0.2883

Table 6: Monolingual goldfish models (1000MB). All models are in Latin script except for the Greek model (Greek script).

	multiblimp	blimp-NL	multinli	snli	sib200	bmlama	truthfulqa	xcomps	arc-easy
EN-NL (tested on English)	0.8416	-	0.3167	0.3289	0.1961	0.3783	0.3418	-	0.2721
EN-NL (tested on Dutch)	0.9725	0.4664	0.3590	0.3567	0.1961	0.4460	0.335	0.5688	0.2819
EN-ES (tested on English)	0.7766	-	0.3167	0.3289	0.1961	0.3546	0.3163	-	0.2751
EN-ES (tested on Spanish)	0.9516	-	0.3167	0.3289	0.1961	0.4011	0.3265	0.5642	0.2963
EN-EL (tested on English)	0.8494	-	0.3167	0.3289	0.1961	0.3624	0.3486	-	0.2755
EN-EL (tested on Greek)	0.9845	-	0.3248	0.3147	0.1961	0.3012	0.3452	0.5368	0.2688
EN-POL (tested on English)	0.8052	-	0.3167	0.3289	0.1961	0.3221	0.3554	-	0.2878
EN-POL (tested on Polish)	0.9471	-	0.3170	0.3293	0.1961	0.3275	0.3588	-	0.2861
NL-EN (tested on Dutch)	0.7224	-	0.3590	0.3289	0.1961	0.2277	0.2806	0.5175	0.2692
NL-EN (tested on English)	0.9662	-	0.3167	0.3289	0.1961	0.4779	0.3299	-	0.2989
ES-EN (tested on Spanish)	0.7855	-	0.3167	0.3289	0.1961	0.2404	0.284	0.5114	0.2743
ES-EN (tested on English)	0.9662	-	0.3167	0.3289	0.1961	0.4712	0.3486	-	0.3056
EL-EN (tested on Greek)	0.666	-	0.3179	0.3294	0.1961	0.2013	0.2925	0.4893	0.2632
EL-EN (tested on English)	0.965	-	0.3167	0.3289	0.1961	0.4199	0.3622	-	0.2823
POL-EN (tested on Polish)	0.6537	-	0.3161	0.3441	0.1961	0.2166	0.2653	-	0.2543
POL-EN (tested on English)	0.9636	-	0.3167	0.3289	0.1961	0.4392	0.352	-	0.2938

Table 7: Sequential B-GPT models.

	multiblimp	multinli	snli	sib200	bmlama	truthfulqa	xcomps	arc-easy
EN-NL (tested on English)	0.969	0.3167	0.3289	0.1961	0.4915	0.3401	-	0.3006
EN-NL (tested on Dutch)	0.961	0.3590	0.3567	0.1961	0.4141	0.3095	0.5587	0.2802
EN-ES (tested on English)	0.970	0.3167	0.3289	0.1961	0.4889	0.3265	-	0.3022
EN-ES (tested on Spanish)	0.937	0.3167	0.3289	0.1961	0.3878	0.3061	0.5371	0.2938
EN-EL (tested on English)	0.969	0.3167	0.3289	0.1961	0.4751	0.3452	-	0.3078
EN-EL (tested on Greek)	0.973	0.3239	0.3147	0.1961	0.2841	0.3146	0.5120	0.2781
EN-POL (tested on English)	0.974	0.3167	0.3289	0.1961	0.4656	0.3486	-	0.3069
EN-POL (tested on Polish)	0.912	0.3176	0.3271	0.1961	0.3235	0.3469	-	0.2832
NL-EN (tested on Dutch)	0.976	0.3590	0.3567	0.1961	0.4317	0.3197	0.5747	0.2747
NL-EN (tested on English)	0.948	0.3167	0.3289	0.1961	0.4515	0.3605	-	0.2976
ES-EN (tested on Spanish)	0.959	0.3167	0.3289	0.1961	0.4001	0.3061	0.5686	0.2921
ES-EN (tested on English)	0.955	0.3167	0.3289	0.1961	0.4518	0.3282	-	0.2955
EL-EN (tested on Greek)	0.943	0.3178	0.3289	0.1961	0.2947	0.3486	0.5427	0.2794
EL-EN (tested on English)	0.984	0.3167	0.3289	0.1961	0.4072	0.3707	-	0.2849
POL-EN (tested on Polish)	0.956	0.3232	0.3567	0.1961	0.3418	0.3861	-	0.2874
POL-EN (tested on English)	0.952	0.3167	0.3289	0.1961	0.4289	0.3520	-	0.2955

Table 8: Simultaneous B-GPT models.

D.1 BLiMP scores per grammatical phenomenon

Phenomenon	Dutch goldfish	EN_NL simultaneous	EN_NL sequential	NL_EN sequential	NL_EN simultaneous
Overall	0.8773	0.5171	0.4664	0.4921	0.509
Adpositional phrases	1.000	0.695	0.475	0.860	0.930
Adverbial modification	0.97	0.49	0.44	0.620	0.490
Anaphor agreement	0.94	0.545	0.61	0.680	0.465
Argument structure	0.87	0.4175	0.4037	0.410	0.4063
Auxiliaries	0.976	0.606	0.36	0.702	0.468
Binding principle A	0.985	0.74	0.735	0.695	0.0.72
Complementive	0.92	0.632	0.622	0.674	0.676
Crossing dependencies	1.000	0.41	0.5	0.220	0.33
Determiners	0.892	0.0875	0.105	0.3075	0.110
Extraposition	0.987	0.6733	0.55	0.590	0.66
Finite argument clause	0.8	0.5033	0.525	0.4667	0.5067
Infinitival argument clause	0.918	0.7878	0.7089	0.6767	0.7278
Nominalisation	0.935	0.595	0.16	0.70	0.39
Parasitic gaps	0.168	0.0725	0.19	0.1375	0.075
Passive	0.887	0.7275	0.715	0.5525	0.75
Quantifiers	0.99	0.595	0.49	0.475	0.47
R-words	0.995	0.18	0.37	0.4550	0.69
Relativisation	0.753	0.5667	0.4267	0.1	0.2367
Topicalisation	0.805	0.26	0.21	0.1	0.2775
Verb-second	1.000	0.51	0.4	0.36	0.5250
Wh-movement	0.776	0.575	0.65	0.49	0.5825
Wh-movement restrictions	0.733	0.7067	0.615	0.5533	0.71

Table 9: Evaluation of BLiMP-NL (Suijkerbuijk et al., 2025) on Goldfish (Chang et al., 2024) and BGPT (Arnett et al., 2025)

Paradigm	English Goldfish	B-GPT NL_EN sequential	B-GPT NL_EN simultaneous	B-GPT EN_NL sequential	B-GPT EN_NL simultaneous
Overall	0.7795	0.7941	0.7844	0.6397	0.7962
adjunct_island	0.8400	0.8200	0.869	0.758	0.770
anaphor_gender_agreement	0.9720	0.9890	0.991	0.937	0.9940
anaphor_number_agreement	0.9900	0.9910	0.988	0.946	0.9900
animate_subject_passive	0.7810	0.7950	0.789	0.6870	0.7810
animate_subject_trans	0.8900	0.8780	0.864	0.8680	0.8680
causative	0.7130	0.7220	0.702	0.5560	0.7200
complex_NP_island	0.4840	0.5930	0.649	0.4380	0.5290
coordinate_structure_constraint_complex_left_branch	0.5670	0.6490	0.641	0.4380	0.6540
coordinate_structure_constraint_object_extraction	0.7390	0.8100	0.821	0.5680	0.8230
determiner_noun_agreement_1	0.9720	0.9810	0.97	0.9270	0.9770
determiner_noun_agreement_2	0.9370	0.9780	0.963	0.7520	0.9690
determiner_noun_agreement_irregular_1	0.8470	0.8810	0.851	0.7170	0.8630
determiner_noun_agreement_irregular_2	0.8970	0.9400	0.937	0.7090	0.9100
determiner_noun_agreement_with_adj_2	0.9000	0.9380	0.935	0.6640	0.8980
determiner_noun_agreement_with_adj_irregular_1	0.8230	0.8830	0.839	0.7130	0.8350
determiner_noun_agreement_with_adj_irregular_2	0.8870	0.9090	0.901	0.6370	0.8760
determiner_noun_agreement_with_adjective_1	0.9370	0.9510	0.951	0.8170	0.9410
distractor_agreement_relational_noun	0.9390	0.9470	0.938	0.6440	0.9270
distractor_agreement_relative_clause	0.6850	0.7840	0.685	0.4560	0.7320
drop_argument	0.7090	0.7720	0.765	0.7470	0.7500
ellipsis_n_bar_1	0.7970	0.7990	0.754	0.5990	0.8240
ellipsis_n_bar_2	0.8840	0.9020	0.892	0.8370	0.9060
existential_there_object_raising	0.7550	0.7590	0.774	0.8090	0.7600
existential_there_quantifiers_1	0.9870	0.9560	0.975	0.9290	0.9850
existential_there_quantifiers_2	0.1410	0.2220	0.124	0.1350	0.1900
existential_there_subject_raising	0.8600	0.8470	0.846	0.7120	0.8740
expletive_it_object_raising	0.7600	0.7530	0.78	0.7420	0.7790
inchoative	0.5720	0.6380	0.629	0.4910	0.6240
intransitive	0.7440	0.7890	0.775	0.6150	0.7700
irregular_past_participle_adjectives	0.9640	0.9980	0.967	0.8680	0.9810
irregular_past_participle_verbs	0.8460	0.9510	0.945	0.7480	0.9320
irregular_plural_subject_verb_agreement_1	0.9040	0.8980	0.897	0.6300	0.9130
irregular_plural_subject_verb_agreement_2	0.9010	0.9160	0.918	0.7420	0.9010
left_branch_island_echo_question	0.3420	0.3220	0.322	0.3740	0.3090
left_branch_island_simple_question	0.6390	0.7480	0.673	0.3890	0.7020
matrix_question_npi_licensor_present	0.6640	0.5520	0.673	0.2820	0.6700
npi_present_1	0.7080	0.5950	0.671	0.2640	0.6450
npi_present_2	0.7760	0.6810	0.832	0.5570	0.7570
only_npi_licensor_present	0.8140	0.8690	0.91	0.5810	0.9110
only_npi_scope	0.7010	0.8280	0.752	0.1230	0.7490
passive_1	0.8660	0.8760	0.871	0.8160	0.8910
passive_2	0.8450	0.8570	0.84	0.8100	0.8610
principle_A_c_command	0.5780	0.5780	0.565	0.4220	0.5600
principle_A_case_1	1.0000	1.0000	1.0	1.0000	1.0000
principle_A_case_2	0.9660	0.9740	0.94	0.6790	0.9660
principle_A_domain_1	0.9770	0.9660	0.968	0.7710	0.9600
principle_A_domain_2	0.7990	0.7090	0.741	0.7190	0.7690
principle_A_domain_3	0.6720	0.6100	0.635	0.5740	0.6390
principle_A_reconstruction	0.3760	0.3770	0.255	0.1230	0.2740
regular_plural_subject_verb_agreement_1	0.9220	0.9690	0.95	0.7930	0.9640
regular_plural_subject_verb_agreement_2	0.8950	0.9100	0.924	0.7650	0.9300
sentential_negation_npi_licensor_present	0.9940	0.9920	0.998	0.8060	0.9980
sentential_negation_npi_scope	0.5760	0.7000	0.662	0.3370	0.7620
sentential_subject_island	0.3840	0.2730	0.343	0.2660	0.4720
superlative_quantifiers_1	0.9410	0.9250	0.718	0.8860	0.9770
superlative_quantifiers_2	0.7830	0.8010	0.727	0.5480	0.8420
tough_vs_raising_1	0.6710	0.6930	0.623	0.4320	0.6550
tough_vs_raising_2	0.8100	0.8560	0.873	0.6940	0.8830
transitive	0.8170	0.8540	0.822	0.6890	0.8360
wh_island	0.8230	0.8390	0.771	0.6380	0.7940
wh_questions_object_gap	0.8020	0.7630	0.746	0.6400	0.8150
wh_questions_subject_gap	0.9300	0.9380	0.951	0.9390	0.9560
wh_questions_subject_gap_long_distance	0.8960	0.8950	0.921	0.9600	0.9060
wh_vs_that_no_gap	0.9590	0.9690	0.978	0.9450	0.9710
wh_vs_that_no_gap_long_distance	0.9790	0.9750	0.989	0.9870	0.9820
wh_vs_that_with_gap	0.4510	0.4490	0.394	0.1810	0.4130
wh_vs_that_with_gap_long_distance	0.2470	0.2240	0.193	0.0330	0.1730

Table 10: BLiMP (en) per grammatical phenomenon. Example data for Dutch models.

E Pairwise Comparison Results

Comparison	Holm-Bonferroni p
English goldfish vs. NL-EN sequential	.051
English goldfish vs. NL-EN simultaneous	1
English goldfish vs. EN-NL sequential	<.001
English goldfish vs. EN-NL simultaneous	<.05
English goldfish vs. ES-EN sequential	.37
English goldfish vs. ES-EN simultaneous	1
English goldfish vs. EN-ES sequential	<.001
English goldfish vs. EN-ES simultaneous	.1
English goldfish vs. EL-EN sequential	1
English goldfish vs. EL-EN simultaneous	<.05
English goldfish vs. EN-EL sequential	<.001
English goldfish vs. EN-EL simultaneous	.5
English goldfish vs. PL-EN sequential	1
English goldfish vs. PL-EN simultaneous	.98
English goldfish vs. EN-PL sequential	<.001
English goldfish vs. EN-PL simultaneous	1

Table 11: Pairwise Wilcoxon signed-rank tests between monolingual Goldfish model and bilingual B-GPT models on English BLiMP, with Holm-Bonferroni corrected p-values.

Comparison	Holm-Bonferroni p
Dutch goldfish vs. EN-NL simultaneous	<.001
Dutch goldfish vs. EN-NL sequential	<.001
Dutch goldfish vs. NL-EN sequential	<.001
Dutch goldfish vs. NL-EN simultaneous	1
EN-NL simultaneous vs. EN-NL sequential	.452
NL-EN simultaneous vs. NL-EN sequential	<.001

Table 12: Pairwise Wilcoxon signed-rank tests between models on Dutch BLiMP, with Holm-Bonferroni corrected p-values.